

Dynamic Regression Models

Jerome Dumortier

Required packages:

- `forecast`
- `Hmisc`
- `stargazer`

Dynamic regression and time series topics:

- Trends and seasonality
- Distributed-lag models (including past or lagged independent variables), e.g.,

$$y_t = \alpha + \beta_0 \cdot x_t + \beta_1 \cdot x_{t-1} + \beta_2 \cdot x_{t-2} + \epsilon$$

- Autoregressive model relating present value of a time series to past values and errors (univariate time series), e.g.,

$$y_t = \beta_0 + \beta_1 \cdot y_{t-1} + \epsilon$$

- Forecasting

Decomposition of data over time into three components:

- Trend
- Season
- Random component

Trend

- Linear time trend: $y_t = \beta_0 + \beta_1 \cdot t + \epsilon_t$
- Exponential time trend: $\ln(y_t) = \beta_0 + \beta_1 \cdot t + \epsilon_t$
- β_1 in the exponential time trend model is the average annual growth rate (assuming t is in years)

Inclusion of a seasonal component via (quarterly in this case) dummy variables:

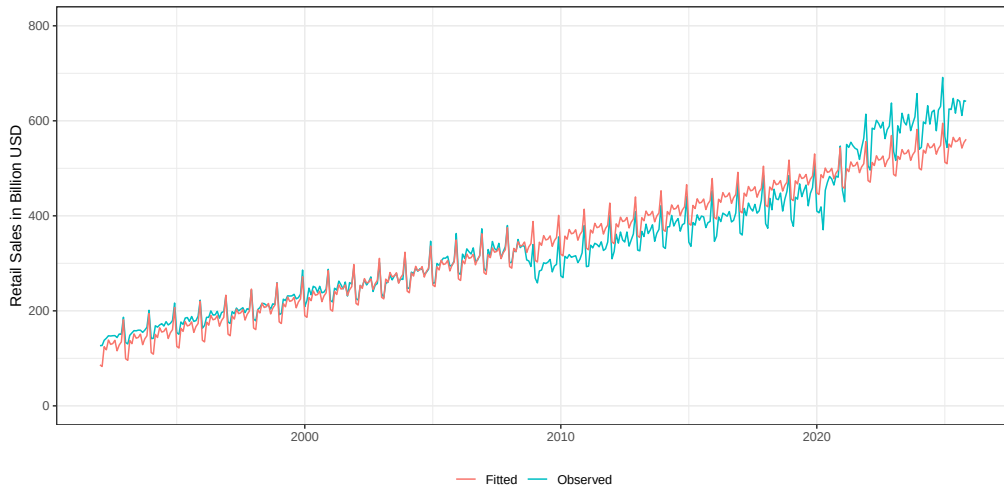
$$y_t = \beta_0 + \delta_1 \cdot Q1_t + \delta_2 \cdot Q2_t + \delta_3 \cdot Q3_t + \beta_1 \cdot x_{1,t} + \cdots + \beta_k \cdot x_{k,t} + \epsilon_t$$

Example using retail Data

Implementation to obtain fitted value:

```
retail$date      = as.Date(retail$date,format="%Y-%m-%d")
retail$month     = months(retail$date)
retail$trend     = 1:nrow(retail)
bhat             = lm(retail~factor(month)+trend,data=retail)
retail$fit       = bhat$fitted.values
```

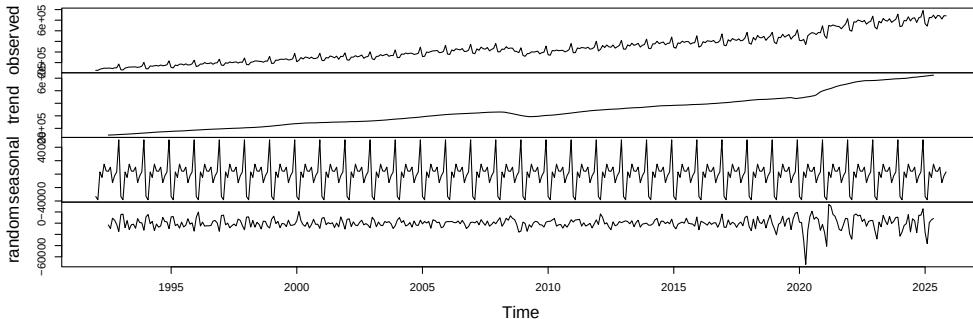
Observed and Fitted retail Data



Decomposition of Time Series

```
retail = ts(retail$retail,start=c(1992,1),frequency=12)  
plot(decompose(retail,type=c("additive")))
```

Decomposition of additive time series



Reasons to Include Lags

Psychological reasons

- Force of habit, e.g., lag in changing consumption habits
- Uncertainty about permanence of change, e.g., getting a new job but with a probationary period.

Technological or economic reasons

- Difficulty to change practices due to high cost

Institutional reasons

- Contractual obligations that cannot be modified in the short-run

Relationship between income and consumption

Assume the following relationship between income and consumption:

$$C_t = \alpha + \beta_0 \cdot I_t + \beta_1 \cdot I_{t-1} + \beta_2 \cdot I_{t-2}$$

Example: Increase in income from \$4,000 to \$5,000

- Assume that $\alpha_0 = 100$, $\beta_0 = 0.4$, $\beta_1 = 0.3$, and $\beta_2 = 0.2$.
- What is the long-run consumption with \$4,000?
- How does the consumption change over the time when receiving the increase of \$1,000

Note that $\sum_{i=0}^2 \beta_i = 0.9$

Long-Run Multiplier

Distributed-lag models (including pasted or lagged independent variables):

$$y_t = \alpha + \beta_0 \cdot x_t + \beta_1 \cdot x_{t-1} + \beta_2 \cdot x_{t-2} + \cdots + \beta_k \cdot x_{t-k} + \epsilon$$

Long-run multiplier (or long-run propensity):

$$\sum_{i=1}^k \beta_i = \beta_0 + \beta_1 + \beta_2 + \cdots + \beta_k = \beta$$

Koyck Method for Distributed-Lag Models

Assumption: All β_k are of the same sign, then $\beta_k = \beta_0 \cdot \lambda^k$ for $k = 0, 1, 2, \dots, \infty$.

Characteristics of this assumption:

- $\lambda < 1$ gives less weight to distant β s
- Long-run multiplier is finite, i.e.,

$$\sum_{k=0}^{\infty} \beta_k = \beta_0 \left(\frac{1}{1 - \lambda} \right)$$

Regression model equation

$$y_t = \alpha + \beta_0 \cdot x_t + \beta_0 \cdot \lambda \cdot x_{t-1} + \beta_0 \cdot \lambda^2 \cdot x_{t-2} + \dots + \epsilon_t$$

Reformulated equation: $y_t = \alpha \cdot (1 - \lambda) + \beta_0 \cdot x_t + \lambda \cdot y_{t-1} + v$

Koyck Method for Distributed-Lag Models

```
##  
## Call:  
## lm(formula = consumption ~ income + Lag(consumption), data = usdata)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -4248.3   -74.8    10.9    91.2   2643.5   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)   -34.98753    47.99975   -0.729  0.466605      
## income          0.07023     0.01832    3.834  0.000152 ***   
## Lag(consumption)  0.92723     0.01997   46.424 < 2e-16 ***   
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 327.3 on 311 degrees of freedom  
##   (1 observation deleted due to missingness)  
## Multiple R-squared:  0.9992, Adjusted R-squared:  0.9992   
## F-statistic: 1.9e+05 on 2 and 311 DF,  p-value: < 2.2e-16
```